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We ran several Monte Hall programs to test the probabilities of certain events. I created methods for the original three doors Monte Hall problem, for birthdays in a class, for reshuffles until you get a valid hand, and for rare candies in a prize pile.

For the birthday monte hall problem we are solving for the probability of having the same birthday as someone in our class. I ran this program for a class of 31 and ran it 10,000 times for accuracy. The result was:

```
Probability is: 0.7325 to have the same birthday
```

This is in decimal form, the percentage would be 73.25%

For the three doors problem, I first tested for the probability of winning the car if you did not choose to switch doors. I tested 10,000 times and printed the result of each test so this screenshot is a small part of those test with the calculated probability at the bottom:

Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Car
Result of door picked with no change:	Goat
Result of door picked with no change:	Car
Result of door picked with no change:	Car
Result of door picked with no change:	Goat
Result of door picked with no change:	Car
Result of door picked with no change:	Goat
Result of door picked with no change:	Car
Result of door picked with no change:	Car
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Car
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Car
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Car
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Result of door picked with no change:	Goat
Probability of wins was:	0.3293

This is in decimal form, the percentage is 32.93%

I then tested for the probability of winning the car if you chose to switch doors every time. This was also run 10,000 times and the result was printed every time. This screenshot shows a small part of those results with the probability at the bottom:

```
Result of final door picked after change: Goat
Original door picked: 2
Result of door number 0: Goat
Result of final door picked after change: Goat
Original door picked: 0
Result of door number 1: Goat
Result of final door picked after change: Car
Original door picked: 1
Result of door number 0: Goat
Result of final door picked after change: Car
Original door picked: 2
Result of door number 0: Goat
Result of final door picked after change: Goat
Original door picked: 2
Result of door number 0: Goat
Result of final door picked after change: Goat
Original door picked: 0
Result of door number 1: Goat
Result of final door picked after change: Car
Original door picked: 1
Result of door number 0: Goat
Result of final door picked after change: Car
Original door picked: 2
Result of door number 0: Goat
Result of final door picked after change: Goat
Original door picked: 1
Result of door number 0: Goat
Result of final door picked after change: Car
Original door picked: 2
Result of door number 0: Goat
Result of final door picked after change: Goat
Probability of wins was: 0.6698
```

This is in decimal form, the percentage is 66.98%

These are the answers to the questions in the book page 34 example 2.20

```
//a- 1/3
//b1- 33%
//b2- 0/dud everytime
//b3- 100% winning chance
//b4- 66%
//b5- switching to the other curtain
```

I also created two monte hall programs for Pokemon tcg. One of them checks the probability of getting a valid hand based on different numbers of pokemon in the deck. For each value of pokemons in the deck the test was run 10,000 times and the probability was printed based on those results:

```
Probability is: 0.1086 for 1 Pokemon
Probability is: 0.2207 for 2 Pokemon
Probability is: 0.3185 for 3 Pokemon
Probability is: 0.3973 for 4 Pokemon
Probability is: 0.4827 for 5 Pokemon
Probability is: 0.5462 for 6 Pokemon
Probability is: 0.5992 for 7 Pokemon
Probability is: 0.6526 for 8 Pokemon
Probability is: 0.6931 for 9 Pokemon
Probability is: 0.7435 for 10 Pokemon
Probability is: 0.7817 for 11 Pokemon
Probability is: 0.8081 for 12 Pokemon
Probability is: 0.8419 for 13 Pokemon
Probability is: 0.8618 for 14 Pokemon
Probability is: 0.8823 for 15 Pokemon
Probability is: 0.8986 for 16 Pokemon
Probability is: 0.9133 for 17 Pokemon
Probability is: 0.9277 for 18 Pokemon
Probability is: 0.9425 for 19 Pokemon
Probability is: 0.9527 for 20 Pokemon
Probability is: 0.9626 for 21 Pokemon
Probability is: 0.9698 for 22 Pokemon
Probability is: 0.9729 for 23 Pokemon
Probability is: 0.9766 for 24 Pokemon
Probability is: 0.983 for 25 Pokemon
Probability is: 0.9862 for 26 Pokemon
Probability is: 0.9898 for 27 Pokemon
Probability is: 0.9904 for 28 Pokemon
Probability is: 0.9925 for 29 Pokemon
Probability is: 0.9951 for 30 Pokemon
Probability is: 0.9959 for 31 Pokemon
Probability is: 0.9961 for 32 Pokemon
Probability is: 0.9977 for 33 Pokemon
Probability is: 0.9972 for 34 Pokemon
Probability is: 0.9994 for 35 Pokemon
Probability is: 0.999 for 36 Pokemon
Probability is: 0.9992 for 37 Pokemon
Probability is: 1.0 for 38 Pokemon
```

Continued:

```
Probability is: 0.9996 for 39 Pokemon
Probability is: 0.9997 for 40 Pokemon
Probability is: 1.0 for 41 Pokemon
Probability is: 1.0 for 42 Pokemon
Probability is: 1.0 for 43 Pokemon
Probability is: 1.0 for 44 Pokemon
Probability is: 1.0 for 45 Pokemon
Probability is: 1.0 for 46 Pokemon
Probability is: 1.0 for 47 Pokemon
Probability is: 1.0 for 48 Pokemon
Probability is: 1.0 for 49 Pokemon
Probability is: 1.0 for 50 Pokemon
Probability is: 1.0 for 51 Pokemon
Probability is: 1.0 for 52 Pokemon
Probability is: 1.0 for 53 Pokemon
Probability is: 1.0 for 54 Pokemon
Probability is: 1.0 for 55 Pokemon
Probability is: 1.0 for 56 Pokemon
Probability is: 1.0 for 57 Pokemon
Probability is: 1.0 for 58 Pokemon
Probability is: 1.0 for 59 Pokemon
Probability is: 1.0 for 60 Pokemon
```

Another Monte Hall program was calculating the probability of being bricked based on different numbers of Rare Candy in the deck from 1-4. Each probability was calculated from running each test 10,000 times

Number of Rare Candies in a Deck of 60	Chance (%) of Bricking
1	90.03999999999999%
2	80.52%
3	72.56%
4	65.06%